Clique and chromatic numbers of the integer-distance graph on planar point sets in general position

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Abstract

Let $A \subset \mathbb{R}^2$ be an infinite set in *general position* (no three points collinear, no four concyclic), and let G_A be the graph on A joining pairs of points at integer Euclidean distance. We study how large the clique number $\omega(G_A)$ and the chromatic number $\chi(G_A)$ can be. We prove, with complete and self-contained arguments: (1) every clique of G_A is finite, via a quantitative self-contained proof of the Anning–Erdős theorem; (2) $\chi(G_A) \leq \aleph_0$ for every admissible A, of any cardinality; (3) an exact transfer principle: some admissible A has $\chi(G_A) = \aleph_0$ if and only if finite admissible configurations of unbounded chromatic number exist, and likewise for clique sizes, where the construction direction rests on a generic rigid-motion placement lemma proved here; and (4) explicit, exactly verified certificates: a six-point set in general position with all fifteen pairwise distances integral, giving $\omega, \chi \geq 6$. Whether χ can equal $\aleph_0$ is thereby shown to be *equivalent* to the open Erdős problem of whether integral point sets in general position exist at every finite size; we do not claim to resolve that problem in either direction. An appendix documents the exact-arithmetic verification.

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1 Statement of the problem and summary of results

Definition 1.1. A set $A \subseteq \mathbb{R}^2$ is *admissible* (in *general position*) if no 3 points of A lie on a line and no 4 points of A lie on a circle. For admissible A, the *integer-distance graph* G_A has vertex set

A and an edge $\{u, v\}$ whenever $|u - v| \in \mathbb{Z}_{>0}$. A *clique* is a subset of A with all pairwise distances positive integers; $\omega(G_A)$ denotes the supremum of clique sizes and $\chi(G_A)$ the least cardinal κ for which a proper κ-colouring exists.

The questions under study: for infinite admissible A, how large can $\omega(G_A)$ and $\chi(G_A)$ be — and in particular, can $\chi(G_A)$ be infinite?

Summary of what is proved and what is open.

(i) Every clique of G_A is finite, for every admissible A (Theorem 2.1 and Corollary 2.2). An infinite clique is impossible.

(ii) There is an explicit six-point admissible set all of whose $\binom{6}{2} = 15$ pairwise distances are integers (Theorem 3.2); consequently $\omega(G_A) \geq 6$ and $\chi(G_A) \geq 6$ for suitable infinite admissible A (Lemma 3.4).

(iii) $\chi(G_A) \leq \aleph_0$ for every admissible A, even uncountable ones (Theorem 4.1). Hence "χ infinite" can only mean $\chi = \aleph_0$.

(iv) Some infinite admissible A satisfies $\chi(G_A) = \aleph_0$ *if and only if* for every n there is a finite admissible configuration of chromatic number at least n (Theorem 5.1); the analogous equivalence holds for clique sizes. In particular, since $\chi \geq \omega$, the existence of integral point sets in general position of every finite size would already force $\sup_A \chi(G_A) = \aleph_0$.

(v) Whether such finite configurations exist at every size is precisely an open problem of Erdős (the largest known integral point set in general position has 7 points; see Section 6). We do not claim a resolution, and no argument below assumes one in either direction.

Thus the unconditional state of knowledge established here is

$$6 \leq \sup_{A \text{ admissible}} \chi(G_A) \leq \aleph_0, \qquad 6 \leq \sup_{A \text{ admissible}} \omega(G_A),$$

every individual clique is finite, and the question of whether the suprema equal $\aleph_0$ is *equivalent* to an open finite combinatorial problem.

2 Every clique is finite

Theorem 2.1 (Anning–Erdős; self-contained quantitative proof). *If $S \subset \mathbb{R}^2$ is infinite and all pairwise distances between points of S are integers, then S lies on a line. Quantitatively, if $A, B, C \in S$ are not collinear, then*

$$|S| \leq 16\big(\lfloor |AB| \rfloor + 1\big)\big(\lfloor |AC| \rfloor + 1\big).$$

Corollary 2.2. *If A is admissible, every clique of G_A is finite.*

Proof of the corollary. An infinite clique would be an infinite set with all pairwise distances integral, hence collinear by Theorem 2.1; but admissible sets contain no three collinear points. $\square$

Proof of Theorem 2.1. Suppose $A, B, C \in S$ are not collinear; in particular they are distinct and $|AB|, |AC| > 0$. Fix any $P \in S$. By the triangle inequality, $k := \big||PA| - |PB|\big|$ is an integer with $0 \leq k \leq |AB|$, and $m := \big||PA| - |PC|\big|$ is an integer with $0 \leq m \leq |AC|$. Hence every point of S lies in

$$L_k(A, B) \cap L_m(A, C), \qquad L_k(A, B) := \big\{X \in \mathbb{R}^2 : \big||XA| - |XB|\big| = k\big\},$$

for one of at most $(\lfloor |AB| \rfloor + 1)(\lfloor |AC| \rfloor + 1)$ pairs (k, m) of integers. It suffices to show that each such intersection contains at most 16 points.

Step 1: algebraic description of the loci. Write $f(X) = |XA|^2$ and $g(X) = |XB|^2$, quadratic polynomials in $X = (x, y)$ whose $x^2 + y^2$ parts coincide; hence $f - g$ is affine, and nonzero because $A \neq B$. For an integer k with $0 < k \leq |AB|$, squaring $\left|\sqrt{f} - \sqrt{g}\right| = k$ twice yields the polynomial condition

$$q_k(X) \;:=\; (f-g)^2 - 2k^2(f+g) + k^4 \;=\; 0, \tag{1}$$

and, conversely, $q_k(X) = 0$ holds if and only if $\big||XA| - |XB|\big| = k$ *or* $|XA| + |XB| = k$. Since always $|XA| + |XB| \geq |AB| \geq k$, the second alternative is vacuous when $k < |AB|$ and contributes exactly the segment $[A, B]$ when $k = |AB|$. Therefore:

- For $0 < k < |AB|$: $Z(q_k) = L_k(A, B)$ is the hyperbola with foci A, B. The degree-2 homogeneous part of q_k is $4\big((A - B) \cdot X\big)^2 - 4k^2|X|^2$, an indefinite, hence nonzero, quadratic form; so $\deg q_k = 2$.
- For $k = |AB|$ (relevant only if $|AB| \in \mathbb{Z}$): $Z(q_k)$ is the union of the two outer rays $\{\, ||XA| - |XB|| = |AB| \,\}$ and the segment $\{\, |XA| + |XB| = |AB| \,\}$, i.e. the *entire line* ℓ_{AB} through A and B. A real quadratic polynomial whose zero set is exactly one line must be a nonzero scalar multiple of the square of a real linear form: a product of two non-proportional real linear forms has a two-line zero set, while a product of complex-conjugate linear forms has real zero set a single point or the empty set. Hence $q_{|AB|} = c\,\ell_{AB}^2$ for a linear form ℓ_{AB} vanishing on the line AB.
- For $k = 0$: take instead $q_0 := f - g$, an affine polynomial whose zero set is the perpendicular bisector of AB.

In every case $L_k(A, B) \subseteq Z(q_k)$ with q_k a nonzero polynomial of degree at most 2. Set $p := q_k(A, B; \cdot)$ and $q := q_m(A, C; \cdot)$, the corresponding polynomials for the pairs (A, B) and (A, C).

Step 2: a hyperbola meets every line in at most two points and is irreducible over $\mathbb{R}$. In canonical coordinates a hyperbola is $x^2/a^2 - y^2/b^2 = 1$. Substituting an affine parametrization $t \mapsto (x_0 + ut,\, y_0 + vt)$, $(u, v) \neq (0, 0)$, of an arbitrary line gives a polynomial of degree ≤ 2 in t with leading coefficient $u^2/a^2 - v^2/b^2$. If this is nonzero there are at most two intersections. If it vanishes, then $v = \pm(b/a)u$ with $u \neq 0$, and the coefficient of t equals $2(x_0u/a^2 - y_0v/b^2)$; were that to vanish as well, we would get $x_0/a = \pm y_0/b$, making the constant term $x_0^2/a^2 - y_0^2/b^2 - 1 = -1 \neq 0$. So the polynomial in t is never identically zero, and *a line meets a hyperbola in at most 2 points*; in particular no line is contained in it. Consequently, for $0 < k < |AB|$ the polynomial q_k admits no real linear factor (such a factor ℓ would force the full line $Z(\ell) \subseteq Z(q_k)$), nor a factorization into complex-conjugate linear forms (whose real zero set is at most one point, not an infinite hyperbola). Hence these q_k are irreducible over $\mathbb{R}$.

Step 3: p and q have no common nonconstant factor. A nonconstant factor of a polynomial of degree ≤ 2 has degree 1 or 2.

Degree 1. A common real linear factor ℓ would give a line $Z(\ell) \subseteq Z(p) \cap Z(q)$. By Step 2 neither p nor q can then be a mid-range hyperbola polynomial, so each is the perpendicular-bisector polynomial ($k = 0$, resp. $m = 0$) or the focal-line polynomial ($k = |AB|$, resp. $m = |AC|$), and $Z(p)$, $Z(q)$ are the corresponding lines. We check all four coincidences of lines.

- *Bisector of AB = bisector of AC.* These lines are perpendicular to AB and to AC respectively, so equality forces $AB \parallel AC$, i.e. A, B, C collinear: excluded.

- *Line AB = line AC.* Then A, B, C are collinear: excluded.
- *Bisector of AB = line AC.* The line AC passes through A; but A lies on the bisector of AB iff $|AA| = |AB|$, i.e. $|AB| = 0$: impossible. Symmetrically for *bisector of AC = line AB.*

Hence no common linear factor exists.

Degree 2. Then p and q are proportional, so $Z(p) = Z(q) =: H$ as sets. If either is a double line, proportionality forces the other to define the same line, reducing to the coincidences just excluded; and a double line cannot be proportional to an irreducible hyperbola polynomial. The remaining case is two genuine hyperbolas: $0 < k < |AB|$, $0 < m < |AC|$, and

$$H \;=\; L_k(A,B) \;=\; L_m(A,C).$$

We derive a contradiction by symmetry. The defining condition $\big||XA| - |XB|\big| = k$ is invariant under the point reflection $\sigma_M(X) = A + B - X$ through the midpoint $M = \frac{A+B}{2}$, since $|\sigma_M(X) - A| = |B - X|$ and $|\sigma_M(X) - B| = |A - X|$ merely swap the two distances. Likewise H is invariant under $\sigma_{M'}$ with $M' = \frac{A+C}{2}$, and $M \neq M'$ because $B \neq C$. The composition $\sigma_{M'} \circ \sigma_M$ is the translation τ by $2(M' - M) = C - B \neq 0$, so H is invariant under a nontrivial translation. Pick any $x_0 \in H$ (the hyperbola is nonempty: it contains its vertices). The orbit $\{x_0 + n(C - B) : n \in \mathbb{Z}\}$ is an infinite collinear subset of H, contradicting Step 2's bound of two points per line. Hence no common quadratic factor exists either.

Step 4: bounding $|Z(p) \cap Z(q)|$ by a resultant. The polynomials p, q are nonzero, of degree ≤ 2, and coprime in $\mathbb{R}[x, y]$ by Step 3. Rotate coordinates: under rotation by an angle φ, the coefficient of the top power of y in (the rotated) p equals the value of the top homogeneous form of p at the direction $(-\sin\varphi, \cos\varphi)$. Each top form is a nonzero form of degree ≤ 2, vanishing in at most two directions, so all but finitely many φ make *both* leading y-coefficients nonzero constants simultaneously; fix such coordinates. Let $d_p, d_q \in \{1, 2\}$ be the y-degrees (the value 0 is excluded because the top form does not vanish on the chosen direction). Form the resultant

$$R(x) \;:=\; \operatorname{Res}_y\big(p, q\big) \;\in\; \mathbb{R}[x].$$

Two standard facts are used. *(i)* If $R \equiv 0$, then p and q have a common factor of positive degree in y over the field $\mathbb{R}(x)$ (Euclidean algorithm in $\mathbb{R}(x)[y]$), hence, by Gauss's lemma, a common nonconstant factor in $\mathbb{R}[x][y] = \mathbb{R}[x, y]$ — excluded by Step 3; so $R \not\equiv 0$. *(ii)* Because the leading y-coefficients are nonvanishing *constants*, specialization commutes with the resultant: for every $x_0 \in \mathbb{R}$,

$$R(x_0) \;=\; \operatorname{Res}\big(p(x_0, \cdot),\, q(x_0, \cdot)\big),$$

which vanishes whenever the univariate polynomials $p(x_0, \cdot)$ and $q(x_0, \cdot)$ share a root. The Sylvester matrix has size $d_p + d_q \leq 4$ with entries of degree ≤ 2 in x, so $\deg R \leq 8$. Hence common zeros of p and q have x-coordinate among at most 8 real numbers, and for each such x_0 at most $d_p \leq 2$ values of y satisfy $p(x_0, y) = 0$. Therefore

$$\big|Z(p) \cap Z(q)\big| \;\leq\; 16.$$

Combining the steps: $|S| \leq 16(\lfloor |AB| \rfloor + 1)(\lfloor |AC| \rfloor + 1) < \infty$ whenever S contains a non-collinear triple, which proves the theorem. (The constant 16 is deliberately not optimized; Bézout's theorem would give 4 per pair (k, m), but only finiteness with an explicit elementary bound is needed.) $\square$

3 Explicit certificates: $\omega \geq 6$ and $\chi \geq 6$

All verifications in this section were performed in *exact integer arithmetic* — perfect-square tests via integer square roots, collinearity via 2×2 integer cross products, concyclicity via 4×4 integer determinants — and then re-verified by an independently written second routine. No floating point is involved anywhere. The code is reproduced in Appendix A.

Proposition 3.1 (warm-up, $n = 4$)**.** *The kite* $\{(0,0), (8,0), (4,3), (4,-3)\}$ *is admissible and its six pairwise distances are* $8, 5, 5, 5, 5, 6$. (Non-concyclicity: the circle through the first three points has centre $(4, -\frac{7}{6})$ and squared radius $\frac{625}{36}$, while $(4, -3)$ lies at squared distance $\frac{121}{36}$ from that centre.)

Theorem 3.2. *There exist six points in general position with all fifteen pairwise distances positive integers; the induced integer-distance graph is* K_6*, so finite admissible configurations achieve* $\omega = \chi = 6$. *Explicitly, with* $\kappa := \sqrt{2002}$*,*

$$\begin{aligned} &P_1 = (0,0), && P_2 = (2312,\ 0), && P_3 = (535,\ 30\kappa), \\ &P_4 = (621,\ -30\kappa), && P_5 = (1691,\ 30\kappa), && P_6 = (1777,\ -30\kappa), \end{aligned}$$

with distance matrix

$$\big(|P_iP_j|\big)_{i,j=1}^{6} = \begin{pmatrix} 0 & 2312 & 1445 & 1479 & 2159 & 2227 \\ 2312 & 0 & 2227 & 2159 & 1479 & 1445 \\ 1445 & 2227 & 0 & 2686 & 1156 & 2958 \\ 1479 & 2159 & 2686 & 0 & 2890 & 1156 \\ 2159 & 1479 & 1156 & 2890 & 0 & 2686 \\ 2227 & 1445 & 2958 & 1156 & 2686 & 0 \end{pmatrix}.$$

Verification scheme. Each squared distance has the exact integer form $(\Delta X)^2 + 2002\,(\Delta m)^2$, where the points are written $(X_i,\ m_i\kappa)$ with $X_i, m_i \in \mathbb{Z}$; for instance

$$|P_1P_3|^2 = 535^2 + 2002 \cdot 30^2 = 286{,}225 + 1{,}801{,}800 = 2{,}088{,}025 = 1445^2.$$

All 15 such integers are perfect squares (checked by integer square root). A triple is collinear iff the cross product of difference vectors vanishes; factoring κ out, this is an integer 2×2 determinant in the (X_i, m_i), and all $\binom{6}{3} = 20$ are nonzero. A quadruple lies on a circle iff

$$\det\big(x_i^2 + y_i^2 \quad x_i \quad y_i \quad 1\big)_{i=1}^{4} = 0;$$

substituting $y_i = m_i\kappa$, the first column becomes the integer $X_i^2 + 2002\,m_i^2$ and κ factors out of the third column (scaling one column by a nonzero scalar does not affect vanishing), so the test is an integer 4×4 determinant; all $\binom{6}{4} = 15$ are nonzero. Hence the configuration is admissible and forms a 6-clique. □

Remark 3.3. The fifteen distances share the factor 17; the similar copy scaled by $\frac{1}{17}$ is an integral six-point set in general position of diameter $2958/17 = 174$. A five-point certificate found en route, with *plain integer coordinates*,

$$(0,0),\ (4225,0),\ (3713,2016),\ (2079,528),\ (2975,-3000),$$

has all ten distances integral (gcd 65; diameter 78 after scaling by $\frac{1}{65}$) and was verified by the same exact scheme. No minimality is claimed for either example.

Lemma 3.4 (extension to infinite sets)**.** *Every finite admissible F is contained in an infinite admissible A with $\chi(G_A) \geq \chi(G_F)$ and $\omega(G_A) \geq \omega(G_F)$.*

Proof. Add points one at a time. A new point must avoid the finitely many lines through pairs and circles through (non-collinear) triples of the current finite set: a finite union of Lebesgue-null sets, so suitable points exist; iterate. The union of the increasing chain is admissible because any three or four of its points already lie in some finite stage. Since G_F remains a subgraph (induced or denser, as new points may add edges), the lower bounds on χ and on clique sizes persist. $\square$

Corollary 3.5. *There exist infinite admissible A with $\omega(G_A) \geq 6$ and $\chi(G_A) \geq 6$.*

4 The chromatic number is at most $\aleph_0$

Theorem 4.1. *For every admissible $A \subseteq \mathbb{R}^2$, of any cardinality, $\chi(G_A) \leq \aleph_0$.*

Proof. Fix $v \in A$. Its neighbours lie on the circles of integer radius $n \geq 1$ centred at v, and admissibility forbids four points of A on any one circle; hence each such circle carries at most three points of A, and

$$\deg(v) \ \leq \ 3 \cdot \aleph_0 \ = \ \aleph_0.$$

In a graph all of whose degrees are countable, every connected component is countable: the breadth-first levels from a fixed vertex are countable by induction (each level is a countable union of countable neighbour sets), and the component is the union of the levels over $\mathbb{N}$. Colour each component injectively by $\mathbb{N}$; vertices in distinct components are non-adjacent, so this is a proper colouring with $\aleph_0$ colours.

Choice bookkeeping. The countable unions use the axiom of countable choice, and assembling colourings across (possibly uncountably many) components uses choice as well; both are available in ZFC, and neither is needed when A is countable — which covers the sets constructed in Theorem 5.1. $\square$

Consequently the dichotomy is sharp: for admissible A, either $\chi(G_A)$ is finite or $\chi(G_A) = \aleph_0$; nothing in between and nothing beyond can occur.

5 Exact reduction of the infinite problem to a finite one

Theorem 5.1 (transfer principle)**.** *The following are equivalent.*

(a) There is an infinite admissible A with $\chi(G_A) = \aleph_0$.

(b) For every $n \in \mathbb{N}$ there is a finite admissible F_n with $\chi(G_{F_n}) \geq n$.

The same equivalence holds with chromatic number replaced by clique size; i.e. some admissible A satisfies $\omega(G_A) = \aleph_0$ (as a supremum) iff integral point sets in general position exist at every finite size. Moreover, the construction in (b)⇒(a) *produces a single* countable *admissible A; combined with Corollary 2.2, such an A would have unbounded finite cliques but no infinite clique.*

Proof of (a)⇒(b). We argue the contrapositive via the De Bruijn–Erdős compactness theorem, proved here for completeness. Suppose some n_0 bounds the chromatic number of every finite admissible configuration. Let A be admissible. Every finite $F \subseteq A$ is admissible, and G_F is precisely the subgraph of G_A induced on F; hence every finite subgraph of G_A is n_0-colourable.

Endow $X := \{1, \dots, n_0\}^A$ with the product topology; it is compact Hausdorff as a product of finite discrete spaces (Tychonoff; this uses the Boolean prime ideal theorem in general, while for countable A a direct diagonal argument suffices). For each edge $e = \{u, v\}$ of G_A, the set

$$C_e \ := \ \{c \in X : \ c(u) \neq c(v)\}$$

is closed, being determined by two coordinates. The family $\{C_e\}$ has the finite intersection property: finitely many edges $e_1, \dots, e_r$ span a finite vertex set F; G_F is n_0-colourable; and any extension of such a colouring to all of A (arbitrary values off F) lies in $C_{e_1} \cap \cdots \cap C_{e_r}$. By compactness $\bigcap_e C_e \neq \varnothing$, and any of its members is a proper n_0-colouring of G_A. So $\chi(G_A) \leq n_0$ for *every* admissible A, refuting (a). $\square$

The construction direction rests on a placement lemma.

Lemma 5.2 (generic placement)**.** *Let S and F be finite admissible sets. Then there is a rigid motion $\rho : X \mapsto R_\theta X + t$ (rotation by θ followed by translation by t) such that $S \cup \rho(F)$ is admissible. Since ρ is an isometry, $G_{\rho(F)} \cong G_F$ arises as an induced subgraph of $G_{S\cup\rho(F)}$, and any additional cross edges can only increase the chromatic number and clique sizes.*

Proof. Degeneracies entirely within S, or entirely within $\rho(F)$, are excluded by admissibility of S and of F together with rigidity of ρ (collinearity and concyclicity are isometry-invariant). The *mixed* degeneracies form finitely many conditions; for fixed θ each is a polynomial equation in $t \in \mathbb{R}^2$ of degree at most 2. We show that, for all θ outside a finite bad set, each condition is a *nonzero* polynomial in t. Each nonzero polynomial condition cuts a Lebesgue-null subset of t-space (a line or a conic), so the union of finitely many such sets is not all of $\mathbb{R}^2$, and a good t exists.

Mixed collinear triples. With $a, b \in S$ and one moved point $\rho(p)$, the condition reads

$$(b - a) \times \big(R_\theta p + t - a\big) = 0,$$

affine in t with linear part the nonzero functional $(b - a) \times (\,\cdot\,)$ (here $u \times v := u_1 v_2 - u_2 v_1$). With one point $a \in S$ and two moved points $\rho(p), \rho(q)$, the direction $\rho(q) - \rho(p) = R_\theta(q - p)$ is fixed, and the condition reads

$$R_\theta(q - p) \times \big(a - R_\theta p - t\big) = 0,$$

again affine in t with nonzero linear part since $q \neq p$. Both are nonzero polynomials for *every θ*.

Mixed concyclic quadruples. Use the circle determinant

$$D(P_1, \dots, P_4) \ = \ \det\big(\,|P_i|^2, \ x_i, \ y_i, \ 1\,\big)_{i=1}^4,$$

which vanishes iff the four points are concyclic or collinear.

- *Three from S, one moved.* The three S-points are non-collinear by admissibility, determining a genuine circle with centre O and radius r; the condition $|R_\theta p + t - O|^2 = r^2$ is a quadratic in t whose $|t|^2$-coefficient equals $1 \neq 0$.
- *One from S, three moved.* The circle through $\rho(p), \rho(q), \rho(r)$ has centre $R_\theta O_F + t$ and the θ-independent radius r_F, where O_F, r_F are the circumcentre and circumradius of $\{p, q, r\}$ (non-collinear by admissibility of F); the condition $|a - R_\theta O_F - t|^2 = r_F^2$ again has $|t|^2$-coefficient 1.

- *Two and two.* Let $a, b \in S$ and $\rho(p), \rho(q)$ be the four points. In D, subtract the $\rho(p)$-row from the $\rho(q)$-row; the resulting row is $\big(\text{affine in } t,\ R_\theta(q-p),\ 0\big)$, and $|t|^2$ now occurs only in the first entry of the $\rho(p)$-row. The coefficient of $|t|^2$ in D is therefore the corresponding 3×3 cofactor

$$\det \begin{pmatrix} a_1 & a_2 & 1 \\ b_1 & b_2 & 1 \\ w_1 & w_2 & 0 \end{pmatrix} = \pm (b-a) \times w, \qquad w := R_\theta(q-p),$$

which vanishes only if $R_\theta(q-p) \parallel (b-a)$ — at most two values of $\theta \in [0, 2\pi)$ for each of the finitely many quadruples (a, b, p, q).

Choose θ outside the finite union of bad angles arising from the two-and-two cases (the other cases impose no condition on θ). Then every mixed degeneracy is a nonzero polynomial in t of degree at most 2; pick t outside the finite union of their null zero sets. □

Proof of (b)⇒(a) *in Theorem 5.1.* Given finite admissible F_n with $\chi(G_{F_n}) \geq n$, set $S_1 := F_1$ and inductively $S_{n+1} := S_n \cup \rho_n(F_{n+1})$ with ρ_n as in Lemma 5.2. Each S_n is finite and admissible, and $A := \bigcup_n S_n$ is a countable admissible set, since any three or four of its points lie in some S_n. For each n, G_A contains an induced copy of G_{F_n}, so $\chi(G_A) \geq n$; hence $\chi(G_A) \geq \aleph_0$, and $\chi(G_A) = \aleph_0$ by Theorem 4.1. The clique version is identical: an n-point integral set in general position, pasted rigidly, remains an n-clique of G_A, so $\omega(G_A) \geq n$ for all n, i.e. $\omega(G_A) = \aleph_0$ as a supremum — while every individual clique of G_A is still finite by Corollary 2.2. □

6 What remains open

By Theorems 3.2 and 5.1, and since $\chi \geq \omega$ always, the question *"can $\chi(G_A)$ be infinite?"* is equivalent to:

Open problem (Erdős). Do there exist, for every n, finite admissible configurations of chromatic number at least n — in particular, do n points in general position with all pairwise distances integral exist for every n?

This is open at the time of writing. The following attributions are from memory and were deliberately *not* verified against the literature, in compliance with the ground rules of this investigation (no internet access; all computation local and exact): the collinearity theorem of Section 2 is due to Anning and Erdős (1945); the largest known integral point set in general position has 7 points (Kreisel and Kurz, ca. 2008, diameter 22270); the minimal diameters for $n = 5$ and $n = 6$ are recorded as 73 and 174 — consistently, the scaled copy of the certificate of Theorem 3.2 has diameter exactly 174 (Remark 3.3); and a structure theorem of Greenfeld, Iliopoulou and Peluse (ca. 2024) shows that planar integral-distance sets are contained in a line or a circle up to very few exceptional points, which strengthens Corollary 2.2 quantitatively but decides the open problem in neither direction.

A subtlety that Theorem 5.1 makes precise deserves emphasis: even a positive resolution would give $\omega(G_A) = \aleph_0$ only as an *unattained supremum*. Theorem 2.1 forbids an infinite clique outright, so an infinite chromatic number, if achievable at all, must come from unbounded *finite* structure.

7 Audit

Scope of claims. Proven unconditionally: Theorems 2.1, 3.2, 4.1, 5.1, Lemmas 3.4, 5.2, and their corollaries. Stated as open, and assumed in neither direction by any proof: whether the suprema of ω and χ over admissible sets equal $\aleph_0$.

Theorem 2.1. The squaring step is treated honestly: $Z(q_k)$ in (1) is the union of the locus L_k and the ellipse $\{|XA| + |XB| = k\}$, and the ellipse is shown empty for $k < |AB|$ and absorbed into the line for $k = |AB|$, so no phantom components enter the analysis. The degenerate indices $k = 0$ and $k = |AB|$ are handled by separate polynomials (affine; double line), and the identification $q_{|AB|} = c\,\ell^2$ is derived from the trichotomy of real factorizations of a quadratic by zero-set type. The common-factor analysis is exhaustive because any nonconstant factor of a degree-≤ 2 polynomial has degree 1 or 2, and each of the four line-coincidence cases terminates in "A, B, C collinear" or "$|AB| = 0$". The identical-hyperbola case uses only invariance under two distinct point reflections, that their composition is a nonzero translation, nonemptiness of the hyperbola, and the two-point line-intersection bound of Step 2. The resultant facts invoked are exactly two: $\mathrm{Res} \equiv 0$ implies a common factor (Euclidean algorithm over $\mathbb{R}(x)$ plus Gauss's lemma), and specialization at constant nonzero leading coefficients; the coordinate rotation guaranteeing constant leading y-coefficients excludes only finitely many angles since the top forms are nonzero. The constant 16 is unoptimized by design; only finiteness is used downstream.

Theorem 3.2. All certificates are exact: integer perfect-square tests, integer cross products, and integer 4×4 determinants with $\sqrt{2002}$ factored out of one column (legitimate, since scaling a column by a nonzero scalar preserves vanishing of a determinant). No floating-point arithmetic occurs anywhere. The six-point configuration was produced by the search code of Appendix A and then re-verified by an independently written checker; all $15 + 20 + 15$ conditions pass. No minimality claims are made for any example.

Theorem 4.1. The hypothesis "no four concyclic" is applied to circles centred at a vertex — a special case — so the bound of three neighbours per integer radius is sound. Countable degrees do *not* by themselves bound the chromatic number of an arbitrary uncountable graph; the proof correctly routes through countability of connected components. The use of choice is flagged explicitly and is vacuous for countable A, which covers the sets built in Theorem 5.1.

Theorem 5.1. The forward direction uses De Bruijn–Erdős, proved inline; its Tychonoff step requires the Boolean prime ideal theorem for uncountable A and is choice-free for countable A. The hidden hypothesis — that finite subsets of an admissible set are admissible and induce the correct subgraphs of G_A — is stated and checked. In Lemma 5.2, the mixed degeneracy classes are enumerated exhaustively (2+1 and 1+2 collinear; 3+1, 1+3 and 2+2 circular; pure cases excluded by rigidity); each is shown to be a nonzero polynomial in t, the only θ-dependence arising in the 2+2 case, where the bad angles are finitely many; and the conclusion uses only that a finite union of Lebesgue-null sets is a proper subset of $\mathbb{R}^2$. The pasting construction preserves lower bounds because rigid motions preserve distances exactly — hence integral adjacency — and additional cross edges cannot decrease the chromatic number or clique sizes.

Compliance. No internet access was used at any point. The attributions in Section 6 are memory-based and labelled as such; no proof depends on them.

A Exact verification code

The configurations of Section 3 are stored as pairs $(X_i, m_i) \in \mathbb{Z}^2$ representing the point $(X_i,\ m_i\sqrt{k})$; the six-point certificate has $k = 2002$, the five-point and four-point examples have $k = 1$ (plain integer coordinates). The independent checker below re-verifies the six-point certificate from scratch; the search program that produced it uses the same exact predicates.

```
# Independent exact verification of the 6-point certificate.
# Points are (X, m*sqrt(2002)) with integers X, m.
from math import isqrt
import itertools

k = 2002
P = [(0,0), (2312,0), (535,30), (621,-30), (1691,30), (1777,-30)]
n = len(P)
assert len(set(P)) == n

# (1) all 15 pairwise distances are positive integers
D = [[0]*n for _ in range(n)]
for i, j in itertools.combinations(range(n), 2):
    d2 = (P[i][0]-P[j][0])**2 + k*(P[i][1]-P[j][1])**2
    s = isqrt(d2)
    assert s*s == d2 and s > 0, ("non-integer distance", i, j, d2)
    D[i][j] = D[j][i] = s

# (2) no 3 collinear: cross product of difference vectors; the factor
#     sqrt(k) cancels from the 2x2 determinant in (X, m) coordinates.
for t in itertools.combinations(range(n), 3):
    a, b, c = (P[i] for i in t)
    cr = (b[0]-a[0])*(c[1]-a[1]) - (b[1]-a[1])*(c[0]-a[0])
    assert cr != 0, ("collinear", t)

# (3) no 4 concyclic: circle determinant with rows (x^2+y^2, x, y, 1);
#     with y = m*sqrt(k) the first column is the integer X^2 + k m^2 and
#     sqrt(k) factors out of the third column, so an integer 4x4
#     determinant decides vanishing.
def det4(M):
    s = 0
    for perm in itertools.permutations(range(4)):
        sg = 1
        for i in range(4):
            for j in range(i+1, 4):
                if perm[i] > perm[j]:
                    sg = -sg
        pr = 1
        for i in range(4):
            pr *= M[i][perm[i]]
        s += sg*pr
    return s

for q in itertools.combinations(range(n), 4):
    M = [[P[i][0]**2 + k*P[i][1]**2, P[i][0], P[i][1], 1] for i in q]
    assert det4(M) != 0, ("concyclic", q)

print("all 15 distances integral; 20 triples non-collinear; "
      "15 quadruples non-concyclic")
for row in D:
    print(row)
```

Output: the distance matrix of Theorem 3.2, with all assertions passing. The five-point certificate of Remark 3.3 is verified by the same routine with $k = 1$.